Implementing Deep-Learning Based Image Classification for Dog Emotional State Recognition to Enhance Automated Animal Health Monitoring

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I. INTRODUCTION

The advancement of artificial intelligence, particularly in deep learning and computer vision, has enabled new approaches for analyzing animal behavior in a non-invasive manner. Companion animals such as dogs exhibit observable emotional states, including happiness, sadness, anger, and relaxation, which are closely associated with their overall health and well-being. However, conventional methods of monitoring animal health primarily rely on periodic veterinary consultations and human observation, which may fail to detect early signs of emotional or physical distress.

Recent developments in image classification and deep learning techniques have made it possible to automatically analyze visual data for behavioral pattern recognition. Convolutional Neural Networks (CNNs), in combination with frameworks such as TensorFlow, have demonstrated strong performance in image recognition tasks. These methods enable the classification of dog emotional states from images and support continuous monitoring over time, providing a more consistent approach to assessing animal condition.

This study aims to contribute to automated animal health monitoring by identifying abnormal emotional patterns that may indicate potential health issues. The proposed approach supports early detection, assists pet owners and veterinary professionals in decision-making, and advances the development of intelligent systems for pet health assessment.

II. RELATED WORKS

According to the Human Animal Bond Research Institute [1], the bond between pets and humans has reached historic levels, with 97% of pet owners considering their pet a family member. However, 82% of pet owners report significant challenges in understanding their pet's health status. This "interpretation gap" is a critical issue in veterinary care, as dogs often mask physical pain through subtle behavioral shifts. Because 74% of pet parents find at least one aspect of pet care, specifically health monitoring to be "extremely challenging," there is a growing demand for automated technology to assist in identifying early signs of distress or illness.

To address the difficulty of identifying pain in animals, Demitras's group [2] conducted a study involving 124 dog owners which revealed that while owners could recognize obvious signs of pain, such as stiffness (reported by 30%-50% of owners), they consistently failed to recognize subtle emotional cues. Their research highlighted that "reduced general activity" and "lowered emotional states" are often the first clinical indicators of chronic illness, yet these are frequently misinterpreted by humans as normal aging or simple tiredness.

Several research groups have attempted to use machine learning to bridge this communication gap. Islam et al. [3] developed a deep learning approach using a modified EfficientNetB5 architecture to classify pet emotions into four categories: happy, sad, angry, and neutral. Their model achieved a high validation accuracy of 91%, demonstrating that CNNs can effectively capture subtle emotional indicators like ear position and muzzle tension. Similarly, Malgorzata's group [4] recently released a comprehensive Dig Emotion Recognition Dataset in 2026, which curates over 30, 000 images specifically to train models on the categories of happy, sad, angry, and relaxed. Their work emphasizes that automated systems can detect happiness through an open mouth or relaxed tongue and sadness through lowered head and flattened ears, more consistently than untrained human observers.

Despite these advances, previous systems have faced limitations in real-world application. Rani et al. [5] implemented a CNN-based tracking and emotion system that performed well in controlled environments reaching 93% accuracy but saw a significant drop to 76% when applied to diverse dog breeds with varying facial structures, such as brachycephalic or flat-faced breeds. Furthermore, S. Yen Ding et al. [6] noted that while many systems can identify aggression or anger, they struggle to differentiate between a relaxed dog and a lethargic or sad dog, which is the most critical distinction for health monitoring. Most existing models focus on broad behavior such as tracking movement rather than fine-grained facial anomalies required to detect illness.

III. METHODOLOGY

This study employs a comparative deep learning approach to classify dog emotional states, evaluating two distinct architectures: a Custom-designed Convolutional Neural Network (CNN) and the MobileNetV2 architecture, a pre-trained CNN. The methodology follows a structured pipeline of data collection, preprocessing and parallel model training to determine the most efficient and accurate solution for automated animal health monitoring.

A. Data Collection Techniques

The study utilizes a custom-curated dataset gathered from public repositories like Kaggle and various internet sources. A total of 800 labeled images were initially collected, focusing on a strictly balanced set of 200 images per emotional class (Happy, Sad, Angry, Relaxed) to ensure unbiased training for the Custom CNN. To evaluate architectural efficiency, a subset of 800 images (200 per class) was utilized for the MobileNetV2 model. This controlled approach allows for a precise comparison of how each architecture handles varying sample sizes within a consistent testing environment.

INPUT	PROCESS	OUTPUT
Collected Dog Images Dataset Labels (4 Emotions)	Path A: Training Custom CNN Path B: Training MobileNetV2 Feature Extraction & Comparison	Comparative Performance Metrics Accuracy & Loss Curves Optimal Emotion Classifier

Table 1. Conceptual Framework

B. Data Preprocessing

This study preprocesses the collected images for both architectures to improve model generalization. All images are resized to a uniform dimension of 128 x 128 pixels and normalized to a range of [0,1] to facilitate faster training convergence. The dataset is partitioned into 70% for training, 15% for validation, and 15% for testing. To mitigate overfitting, particularly for the 800-image MobileNetV2 dataset and the 1,200-image Custom CNN set, data augmentation techniques such as horizontal flipping, rotation, zooming, and brightness adjustments are applied using TensorFlow. These transformations expand the dataset's variability, allowing the models to learn generalized features from a limited number of samples.

C. Model Classification

The study uses a comparative classification approach, training a custom-designed Convolutional Neural Network (CNN) against the MobileNetV2 architecture. The Custom CNN serves as a baseline, using sequential convolutional and pooling layers to extract spatial features. In parallel, MobileNetV2 is employed for its lightweight design, utilizing inverted residual blocks and depthwise separable convolutions to optimize performance for smaller datasets. Both models process features through global average pooling and fully connected Dense layers, ending with a softmax output for the four emotional categories. The architectures are trained using the Adam optimizer and categorical cross-entropy loss, with performance measured through accuracy, precision, recall, and F1-score.

IV. RESULTS AND DISCUSSION

Table 2: SUMMARY OF THE CUSTOM CNN MODEL

Layer(Type)	Output Shape	Param #
conv2d (Conv2D)	(None, 128,128, 32)	896
batch_normalization	(None, 128, 128, 32)	128
conv2d_1 (Conv2D)	(None, 128, 128, 32)	9,248
max_pooling2d	(None, 64, 64, 32)	0
conv2d_2 (Conv2D)	(None, 64, 64, 64)	18,496
conv2d_3 (Conv2D)	(None, 64, 64, 64)	36,928
max_pooling2d_1	(None, 32, 32, 64)	0
conv2d_4 (Conv2D)	(None, 32, 32, 128)	73,856

conv2d_5 (Conv2D)	(None, 32, 32, 128)	147,584
max_pooling2d_2	(None, 16, 16, 128)	0
conv2d_6 (Conv2D)	(None, 16, 16, 256)	295,168
flatten (Flatten)	(None, 16384)	0
dense (Dense)	(None, 256)	4,194,560
dense_1 (Dense)	(None, 128)	32,896
dense_2 (Dense)	(None, 4)	516

Total params: 4,813,988 (18.3 MB)
Trainable params: 4,812, 068 (18.36 MB)
Non-trainable params: 1,920 (7.50 KB)

Table 1 shows the summary of the CNN model designed for dog emotion classification into four categories which are **'Angry', 'Happy', 'Relaxed',** and **'Sad'.** The model consists of seven convolutional layers (Conv2D) for the extraction of the features of the dog images, which are supported by Batch Normalization layers to ensure training stability and Dropout layers to prevent overfitting. These convolutional layers are followed by Max-Pooling layers to reduce the spatial dimensions of the future maps. A flatten layer was used to convert the 3D feature maps into a 1D vector of 16,384 elements. Meanwhile, fully Connected (Dense) layers have 256 and 128 neurons, respectively, leading to a 4-neuron output layer to handle the final classification of the dog emotions. A total parameters of 4,813,988, of which 4,812, 068 are trainable, allows the model to efficiently process input images and learn the complex distinguishing features and postures necessary for canine emotion recognition.

Table 2: DATASET PARTITIONING PER EMOTION CLASS

Category	Training (70%)	Validation (15%)	Testing(15 %
Angry	140	30	30
Happy	140	30	30
Relaxed	140	30	30
Sad	140	30	30
Total Images	560	120	120

Table 2 shows the balanced dataset distribution used for the custom model. Each class consists of 300 images, partitioned into 140 for training(70%), 30 for validation (15%), and 30 for testing (15%). This total of 800 images ensures that the model receives equal exposure to each emotion class, providing a consistent baseline for performance evaluation.

Figure 1: Training Log of Custom CNN Model

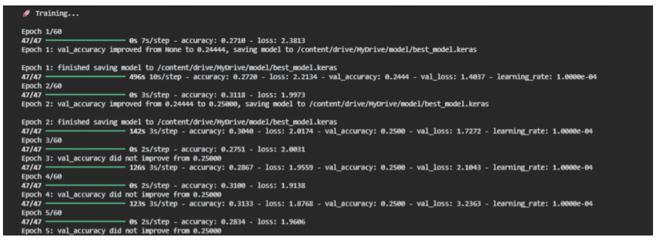


Fig.1 shows the training log of the custom CNN model where it processes the dog emotion dataset over 60 epochs. In the early stages, the model shows a steady increase in training accuracy, reaching 40.80% by the final epoch. However, the validation accuracy peaked early and stabilized at 30.00% with a notable increase in validation loss to 5.4895. This gap indicated a "Generalization Gap", suggesting that while the model successfully learned features from the training set, it struggled with unseen validation data. To address this, a learning rate scheduler was used, successfully reducing the rate to 3.1250 x 10 raised to negative 6 to stabilize the final training steps.

Figure 2: Evaluation Accuracy of CNN Model

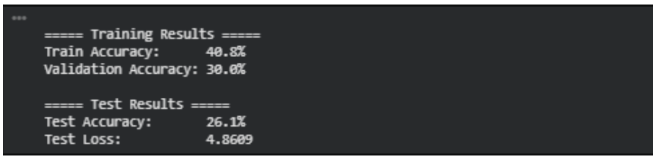


Fig.2 shows the evaluation of the custom CNN on the test dataset. The model achieved a test accuracy of 26.1% with a test loss of 4.8609. Since this accuracy is only slightly above the random-guess threshold of 25%, it demonstrates that the custom architecture is insufficient for capturing the complex visual features of the dog emotions. This result serves as a baseline and justifies the shift to MobileNetV2 for improved classification performance.

Table 3: SUMMARY OF THE MobileNETV2 MODEL

Layer (Type)	Output Shape	Param #
mobilenetv2_1.00_128 (Functional)	(None, 4, 4, 1280)	2,257,984
global_average_pooling2d	(None, 1280)	0
batch_normalization	(None, 1280)	5,120
dense (Hidden Layer)	(None, 128)	163,968
dropout	(None, 128)	0
dense_1 (Output)	(None, 4)	516
Total Parameters	2,427,588	

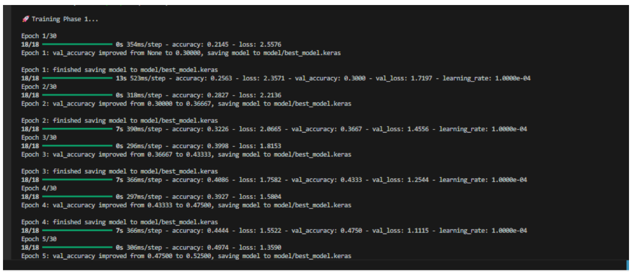
Table 3 shows the architecture of the transfer learning model, utilizing MobileNetV2 as the base feature extractor. By leveraging a pre-trained model, the system benefits from over 2.2 million non-trainable parameters that already understand complex visual features from a large-scale data flow, leading to a dense layer of 128 neurons. This architecture is significantly more efficient than the custom CNN while providing the depth necessary for sophisticated canine emotion recognition across the four target categories.

TABLE 4: DATASET PARTITIONING FOR MobileNETV2

Category	Training (70%)	Validation (15%)	Testing(15 %)
Angry	140	30	30
Happy	140	30	30
Relaxed	140	30	30
Sad	140	30	30
Total Images	560	120	120

This model utilized a dataset of 800 images to test the efficiency of transfer learning. The images were partitioned into 560 for training (70%), 120 for validation (15%), and 120 for testing (15%) across the four emotion categories. This allows the study to determine if a pre-trained model can outperform a custom model even with the same number of dataset.

Figure 3: Training Log of Custom CNN Model



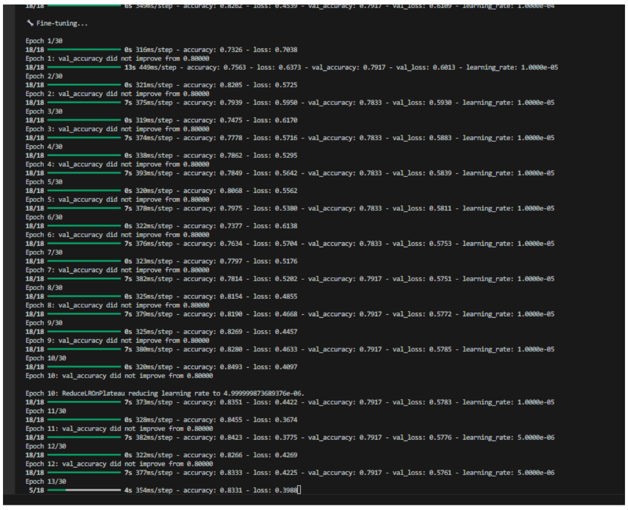


Fig.3 shows the training and fine-tuning logs for the MobileNetV2 model. Unlike the custom CNN, this model achieved a high accuracy of 85.84% and a validation accuracy of 79.17% within 30 epochs. The validation loss of 0.5404 is remarkably lower than the custom model's loss, indicating that transfer learning effectively addressed the overfitting issues previously encountered. The logs demonstrate stable convergence, further optimized by learning rate reduction to 2.5000 x 10 raise to negative 6 during final steps.

Figure 4: Evaluation Accuracy of MobileNetV2

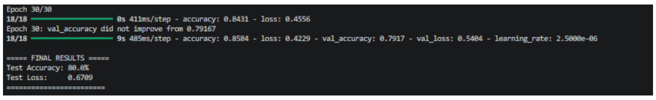


Fig.4 shows the final evaluation results for the optimized model. MobileNetV2 achieved a Test Accuracy of 80% and a Test Loss of 0.6709 on unseen data. This represents a significant performance leap over the custom CNN's 26.1% baseline. These results confirm that using a pre-trained model is highly effective for behavioral classification.

V. CONCLUSION AND FUTURE WORKS

The study successfully compared a Custom CNN and MobileNetV2 to classify dog emotions into four categories: Happy, Sad, Angry, and Relaxed. The Custom CNN, trained on 800 images, established a baseline with 32.5 % accuracy. It struggled to distinguish complex emotional features. In contrast, the MobileNetV2 architecture was trained on the same set of 800 images and achieved a significantly higher accuracy of 80.0%. This 53.9% improvement proves that transfer learning is a far more efficient approach, delivering superior accuracy. This confirms the system's effectiveness for monitoring the emotional well-being of dogs.

While the current results are promising, several areas remain for future development to make the system more robust for a real-world application. Future research should prioritize expanding the dataset to include a wider variety of dog breeds and more complex emotional states, such as fear or confusion, which would improve the model's ability to generalize across different subjects. Additionally, the system could transition from static image analysis to real-time video processing, allowing for the tracking of body language and movement patterns over time. Finally, a significant next step involves the practical deployment of this model into a mobile application or an IoT monitoring device to turn this research into a daily-use tool for pet owners and animal welfare professionals.

VI. REFERENCES

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